Interactive Enhanced Network Based on Multihead Self-Attention and Graph Convolution for Classification of Hyperspectral and LiDAR Data

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***Abstract*— The fusion of multimodal data plays a crucial role in classification tasks. However, existing research typically mines and analyzes the individual features of each data source separately before considering how to fuse them. In contrast, our approach first constructs interactive enhanced fusion features (IEFFs) for initial fusion while considering the extraction of individual features and, finally, integrates them effectively to utilize the information from each data source more comprehensively. To this end, we propose a novel interactive enhanced network based on multihead self-attention (MSA) and graph convolution. Specifically, we extract individual features from hyperspectral image (HSI) and light detection and ranging (LiDAR) data and then construct IEFFs based on the row and column features of the central pixel. Individual features focus on the local characteristics of a single data source, while IEFFs strengthen the feature expression of the central pixel through matrix operations, integrating the complementary information of multimodal data. Subsequently, we use graph convolutional networks (GCNs) to construct graph structures for four types of features (interactive enhanced HSI features, interactive enhanced LiDAR features, HSI individual features, and LiDAR individual features), modeling the pixels as nodes and capturing spatial relationships. On this basis, we apply an MSA mechanism to mine spectral dependencies, further extracting global spectral features. Finally, we design a multimodal gated fusion module (MGFM) that effectively integrates these features through its weighting mechanism. The weight allocation is adjusted dynamically according to the characteristics of the feature, achieving optimal fusion of multimodal data. Extensive experiments on three popular HSI and LiDAR datasets verify the superior performance of our method. Our code will be available at https://github.com/haofeng0003/MSA-GCN.**

***Index Terms*— Graph convolutional network (GCN), hyperspectral images (HSIs), joint classification, light detection and ranging (LiDAR) data, multihead self-attention (MSA).**

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I. Introduction

HYPERSPECTRAL image (HSI), with its rich spectral information, has been widely applied in various fields such as urban land cover classification, target detection [1], [2], and anomaly detection [3], [4]. However, in highly heterogeneous urban areas, the sole use of HSI faces challenges of accuracy and resolution [5]. In contrast, light detection and ranging (LiDAR) data, with its elevation information, can effectively distinguish spectrally similar but materially different objects and is insensitive to atmospheric conditions and lighting changes [6], [7]. By integrating HSI and LiDAR data in a multimodal classification approach, the advantages of both data can be fully utilized, significantly improving the accuracy and reliability of urban land cover classification [8]. HSI provides detailed spectral information, while LiDAR data offer precise geometric information, aiding in more accurately identifying objects' shapes and heights [9]. This method enhances adaptability to complex urban environments, providing more effective solutions for fields such as geological exploration [10], forest management [11], precision agriculture [12], urban planning [13], and environmental monitoring [14].

In recent years, HSI processing technology has rapidly advanced, leading to significant progress in various tasks such as classification, unmixing, and super-resolution. Traditional spatial–spectral classification methods primarily rely on local information and struggle to capture global correlations in discontinuous regions. To address this, a newly proposed normalized spectral clustering method based on kernel learning integrates local and global features to enhance classification performance [15]. In the unsupervised multispectral-aided HSI super-resolution task, a new unmixing-inspired network leverages attention mechanisms to improve super-resolution capabilities [16]. For hyperspectral unmixing, the cyclic consistency unmixing network employs perceptual mechanisms to better preserve material information, thereby enhancing

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unmixing performance [17]. In addition, nonlinear tensor unmixing methods optimize tensor structures to improve unmixing results [18]. In adversarial defense, context-aware networks significantly enhance model robustness through better context modeling [19].

Convolutional neural networks (CNNs), due to outstanding performance in image processing, can effectively extract and learn spatial and spectral features from images, making them widely used in image classification tasks [20], [21], [22]. The multilayer structure of CNNs allows them to progressively extract features from low level to high level, enabling deep learning of complex image information [23], [24]. First, Wang et al. [25] proposed a method that combines shallow and deep convolutional features in the spatial feature extraction process of HSI. By integrating shallow and deep convolutional features, the focus is on more global information, thereby improving classification performance. At the same time, they introduced dilated convolutions in the LiDAR branch to achieve a larger receptive field and reduce information loss. This method effectively combines features from different levels, enhancing the overall performance of the model. In another study, a joint feature learning and fusion mechanism using CNN and spatial morphological blocks was proposed [26]. Spatial morphological blocks can capture information related to the height or shape of different land cover areas from LiDAR data, thereby improving classification accuracy and reliability. Wang et al. [27] proposed a multiscale spatial–spectral cross-modal attention network (MS2CANet), where the multiscale pyramid convolution is used to extract multiscale features and the effective feature recalibration module is used to enhance features at each scale and suppress irrelevant information. Through this multiscale feature extraction and enhancement mechanism, MS2CANet can better adapt to complex HSI.

Gabor texture features have also been introduced into CNNs to enhance the model's adaptability to sample orientation changes [28]. Peng et al. [29] adopted Gabor filters to extract Gabor features, capturing the texture features of samples and, thereby, improving the classification performance of the CNN model. However, traditional Gabor filters may have limitations in multiscale and multidirectional feature extraction. Liu et al. [30] improved convolution by using fractional Gabor filters, enabling the extraction of multiscale and multidirectional features. At different feature extraction stages, the use of multiscale fractional Gabor filters enhances the complementarity of heterogeneous features, effectively improving classification performance.

However, existing methods may lack effective handling of structural information transmission and physical property alignment, thus weakening the complementary capabilities of multisource data in collaborative classification tasks [31]. Zhang et al. [32] utilized a directionwise attention mask derived from LiDAR data to assist in constructing spectrally enhanced feature mappings, optimizing the transmission of HSI information and reducing the uncertainty of HSI data. Fang et al. [33] designed a forward-and-inverse CNN structure as a feature extractor. The spatial enhancement module uses LiDAR features to enhance the spatial representation of HSI, while the spectral enhancement module uses hyperspectral features to enhance the spectral representation of LiDAR data. Lu et al. [34] proposed a coupled adversarial learning-based classification (CALC) method for HSI and LiDAR data fusion classification. By establishing an adversarial game, the learned features can retain the detailed information of HSI and LiDAR data, respectively. Simultaneously, the design of weighted sharing and linear fusion structures in the dual generators allows the extraction of both similar high-order semantic information and modality-specific complementary information.

Due to the inherent differences in attributes and semantic information between heterogeneous data, most existing deep learning methods may not effectively extract features from both data sources when utilizing their interactive information [35], [36]. Sun et al. [37] proposed a multiscale 3-D–2-D mixed CNN feature extraction and multisource data lightweight attention-free fusion network (M2F). This network uses a morphological feature enhancement module and a deep dilation convolution module to effectively extract morphological features and explore the deep spectral–spatial-elevation features of the data. In this way, efficient interaction and integration of multiscale CNN-derived features are achieved, effectively utilizing the complementary advantages provided by various data sources. Wang et al. [38] designed a cross-modal enhanced CNN and Transformer module to achieve effective interaction of information between HSI and LiDAR from both local and global enhancement perspectives. Zhao et al. [39] combined CNN and Transformer networks to design a dual-branch hierarchical network for joint classification of HSI and LiDAR data. The deep hierarchical architecture of this network not only fully utilizes the powerful spatial contextual information extraction capabilities of CNNs but also achieves deep fusion of spectral features in HSI and elevation information in LiDAR based on the self-attention mechanism of Transformer networks. Song et al. [40] designed a Transformer-based spectral encoder and a CNN-based spatial encoder. With the help of dual-structure feature encoders, the long-distance spectral sequence information of HSI and the local spatial information of LiDAR images are effectively encoded. Finally, an edge decoder is built on the HSI features to predict binary LiDAR edge maps, achieving cross-modal information interaction.

In recent years, attention mechanisms have gained widespread attention for improving feature extraction and fusion of multisource data [41], [42]. Traditional CNNs may struggle to fully capture long-range dependencies and salient features in complex multisource data [43]. Attention mechanisms address this issue by dynamically allocating feature weights, thereby enhancing the effectiveness of feature extraction and fusion [44]. Zhang et al. [45] proposed a mutual guidance attention (MGA) module. The MGA module increases the flow of information between the spatial and elevation branches, highlighting essential features while suppressing irrelevant ones. MGA exchanges attention maps between the two branches to emphasize their respective spatial information. Dong et al. [46] embedded an effective self- and cross-guided attention (SCGA) module in their network.

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SCGA not only refines the multimodal features extracted from the HSI and LiDAR streams but also provides cross-guidance from the LiDAR stream to the HSI stream, leveraging the strengths of multimodal data to compensate for the weaknesses of each modality, resulting in a more effective mix of spectral–spatial-elevation information in the network.

Moreover, the multihead self-attention (MSA) mechanism of Transformers captures more global and local feature information by focusing on different subspaces in parallel, significantly enhancing the model's feature representation capability and classification performance [47], [48], [49]. Ding et al. [50] were the first to introduce a Transformer network for the fusion classification of HSI and LiDAR data. By considering spectral features from a sequential perspective, the global–local transformer network (GLT-Net) uses multihead attention to learn global spectral dependencies from grouped local spatial embeddings, effectively extracting and utilizing global spectral sequence features. Zhang et al. [51] designed a local-based multisource feature interactor, providing an environment for information exchange among multibranch features and alleviating the homogeneity processing mode of the self-attention process. Tu et al. [52] proposed a Transformer architecture based on local semantic feature aggregation to enhance multiscale feature learning by using pixel aggregation and local semantic capture, reducing computational costs. Zhang et al. [53] developed a cross-modal feature aggregation and enhancement network for fusing hyperspectral and LiDAR data, improving classification performance through addition and concatenation strategies, reversible neural networks, and feature enhancement modules. Tu et al. [54] introduced a network combining global and local features, using CNNs to extract high-frequency features and a Transformer to capture low-frequency information for effective multisource remote sensing data fusion.

Redundant information in multisource data often affects the effectiveness of heterogeneous feature extraction [55]. To address this issue, researchers have proposed using graph structures to improve feature extraction [56], [57]. Cai et al. [58] designed a graph attention-based multimodal fusion network (GAMF), constructing an undirected weighted graph using Euclidean distance and employing a graph attention strategy to fuse the information of each node, thereby uncovering the relationships between multisource data. However, the Euclidean distance can only reflect the absolute differences between features and fails to capture relative differences. This approach may overlook spectral internal connections, resulting in extracted features that are neither accurate nor comprehensive. To address this problem, Guo et al. [59] proposed a dual graph convolutional network (GCN) for joint dense classification of HSI and LiDAR data. They used the Euclidean distance and cosine similarity to construct graphs, capturing absolute and relative spectral differences. Considering the long-range dependencies in HSI, the rich spectral information in the HSI data cube should be viewed as a series of spatial–spectral signals rather than relying solely on traditional CNNs for local feature extraction. Cai et al. [60] proposed an anchor-based multiview kernel subspace clustering with spatial regularization (AMKSC). By combining kernel methods with spatial regularization, they modeled complex nonlinear correlations and spatial interdependencies between multiple modalities, resulting in a more accurate graph structure. Feng et al. [61] established multibranch dual-channel GCNs to obtain multiview feature representations, uncover potential relationships within the same view, and explore downstream complementarity and consistency between different views. Liao et al. [62] proposed a class-embedded active learning framework that uses a class-level GCN to more effectively capture the structural relationships between samples in HSI classification, thereby optimizing sample selection and labeling efficiency.

Based on the above analysis, this article proposes a novel interactive enhanced network based on MSA and graph convolution. Unlike existing studies, which typically explore and analyze the individual features of each data source separately and then consider how to fuse them, our method first constructs interactive enhanced fusion features (IEFFs) for the initial fusion of multimodal data, obtaining IEFFs with HSI and LiDAR as the main components, respectively. Then, the features are integrated using a GCN and MSA. GCN is used to construct graph structures for individual and IEFFs, modeling pixels as nodes and capturing spatial relationships. Next, MSA is applied to extract global spectral features further. Finally, the features are effectively integrated through a weighted sum using each feature's weight. Extensive experiments on three popular HSI and LiDAR datasets have demonstrated the superior performance of our method.

The main contributions of this article are given as follows.

1) *Extracting Individual Features and Constructing IEFFs:* First, individual features are extracted separately from HSI and LiDAR data. IEFFs are constructed based on the row and column features of the central pixel. Individual features focus on the local characteristics of a single data source, while IEFFs improve the feature expression capability of the central pixel through matrix operations in the row and column directions. This approach fully utilizes the complementary nature of multimodal data and preserves the unique characteristics of each data source at the initial stage of feature extraction.
2) *Utilizing GCN and MSA Mechanism for Feature Extraction:* In the feature extraction stage, GCN is first used to construct graph structures for four types of features (interactive enhanced HSI features, interactive enhanced LiDAR features, individual HSI features, and individual LiDAR features). Each feature is represented by pixels as nodes, which model each pixel and its neighborhood to capture spatial relationships. By constructing an adjacency matrix, spatial dependencies among pixels are revealed. On this basis, MSA is applied to extract global spectral features further. MSA can learn global dependencies in the spectral dimension by processing spectral information in parallel with multiple heads. It focuses on different spectral features in the data, thereby better capturing the correlations among them. This combination effectively integrates spatial and spectral information, improving the model's feature extraction capability and classification performance.

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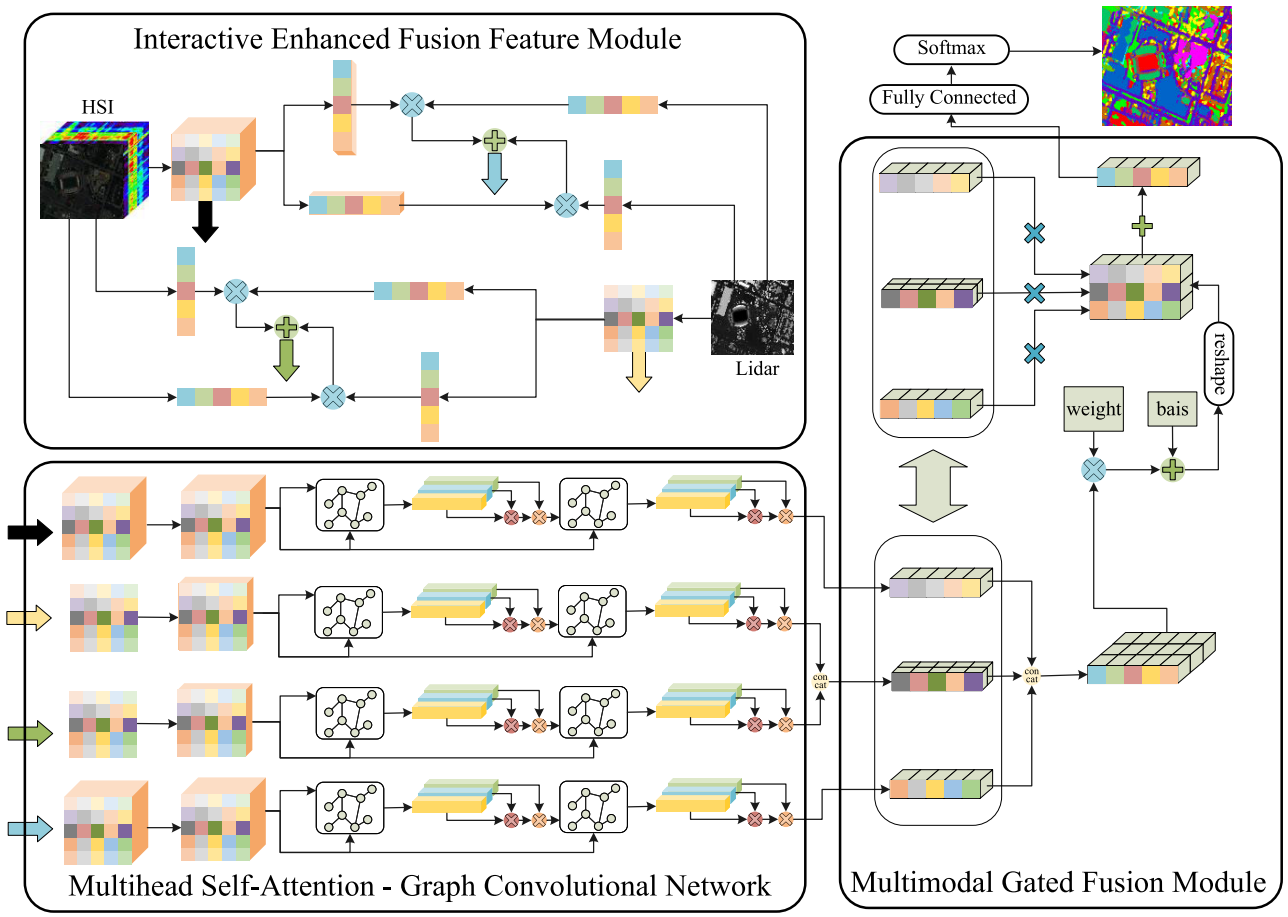


Fig. 1. Proposed framework of the interactive enhanced network based on MSA and graph convolution for classification of hyperspectral and LiDAR data.

3) *Designed a Multimodal Gated Fusion Module (MGFM):* To achieve effective feature fusion, we devised an MGFM. This module utilizes gating mechanisms to weigh and sum features from different data sources and processing modules, combined with original features, to generate comprehensive feature representations. The gating mechanism calculates weight coefficients for each feature, dynamically adjusting weight allocation based on data feature variations, ensuring optimal fusion of multimodal data. This approach not only enhances feature expression but also maximizes the use of complementary information among different features, thereby optimizing the fusion of multimodal data. The module's design significantly enhances the model's adaptability and classification performance with complex multimodal data.

The rest of this article is organized as follows. Section II elaborates on the specific details and principles of the proposed method. Subsequently, Section III provides a detailed presentation and analysis of the experimental results. Finally, Section IV summarizes the research findings and presents conclusive viewpoints.

II. Proposed Approach

The framework of the proposed method for joint classification of HSI and LiDAR data is shown in Fig. 1. The classification method comprises three main components: IEFF module, MSA-graph convolutional network (MSA-GCN), and MGFM. Therefore, we will provide a detailed explanation of each part. Given HSI $\mathcal{H} \in \mathbb{R}^{m\times n\times C}$ and corresponding LiDAR image $\mathcal{L} \in \mathbb{R}^{m\times n\times d}$ covering the same region of Earth's surface, where m and n denote the height and width of the images and C and d denote the number of spectral bands for HSI and LiDAR data, respectively, principal component analysis (PCA) is used to reduce the dimensionality of HSI data, minimizing redundant spectral information, resulting in $\mathfrak{h} \in \mathbb{R}^{m\times n\times c}$. Then, for each pixel in the image, small cubes centered on the pixel are selected from X_h and X_l, denoted as $x_h \in \mathbb{R}^{p\times p\times c}$ and $x_l \in \mathbb{R}^{p\times p\times d}$, respectively.

In existing multimodal data fusion methods, features from HSI and LiDAR data are typically extracted separately and then simply fused, which may not fully leverage the complementary information between these data types. The IEFF module aims to overcome this limitation by integrating and enhancing features at an early stage of processing.

To capture and integrate features of the central pixels, the following processing steps are performed on the HSI and LiDAR data: $x^h_{\text{col}} \in \mathbb{R}^{p\times 1\times c}$ extracts the central pixel columns from x_h, representing data from the central columns extracted across all channels of each HSI cube. $x^h_{\text{row}} \in \mathbb{R}^{1\times p\times c}$ extracts the central pixel rows from x_h, representing data from the central rows extracted across all channels of each HSI cube.

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$x_{\text{col}}^{l} \in \mathbb{R}^{p\times 1\times d}$ extracts the central pixel columns from x_l, representing data from the central columns extracted across all channels of each LiDAR image cube. $x_{\text{row}}^{l} \in \mathbb{R}^{1\times p\times d}$ extracts the central pixel rows from x_l, representing data from the central rows extracted across all channels of each LiDAR image cube.

The cubes x_h and x_l are designed to capture essential information from the central pixels of the images, while the horizontal and vertical cubes x_{col}^{h}, x_{row}^{h}, x_{col}^{l}, and x_{row}^{l} aim to gather information related to the central pixel on the same row or column. This approach effectively utilizes spatial local information of the images and considers features in different directions.

A. Interactive Enhanced Fusion Feature Module

First, dimension reduction operations were performed on the HSI (x_{col}^{h}, x_{row}^{h}) and LiDAR data (x_{col}^{l}, x_{row}^{l}), calculating means along the spectral channels. This operation aimed to compute the mean values for each image channel, resulting in h_{col}, h_{row}, l_{col}, and l_{row}. Two convolution operations were employed when processing the HSI x_h and LiDAR data x_l, followed by batch normalization and rectified linear unit (ReLU) activation function processing of the extracted features. Convolution kernels of sizes $(3, 1)$ and $(1, 3)$ were simultaneously applied to x_h and x_l. These two convolution operations aim to extract spatial features of the image from different directions. By iteratively reducing the spatial dimensions of the features, we obtain $f_{\text{row}}^{h} \in \mathbb{R}^{1\times p\times c}$, $f_{\text{col}}^{h} \in \mathbb{R}^{p\times 1\times c}$, $f_{\text{row}}^{l} \in \mathbb{R}^{1\times p\times d}$, and $f_{\text{col}}^{l} \in \mathbb{R}^{p\times 1\times d}$. This approach better captures the central pixel's subtle changes and important features, thereby increasing the model's attention to the central region.

Next, perform matrix multiplication of the HSI's row feature f_{col}^{h} and column feature f_{row}^{h} with the LiDAR data's x_{row}^{l} and x_{col}^{l}, respectively. Then, batch normalization and ReLU activation are applied, and finally, the square root of the result is taken. The formula can be expressed as

$$f_h = \sqrt{\text{ReLU}\left(\text{BN}\left(f_{\text{col}}^{h} \cdot x_{\text{row}}^{l}\right)\right)} + \sqrt{\text{ReLU}\left(\text{BN}\left(x_{\text{col}}^{l} \cdot f_{\text{row}}^{h}\right)\right)} \quad (1)$$

representing $f_h \in \mathbb{R}^{p\times p\times c}$. The processing for LiDAR data was similar

$$f_l = \sqrt{\text{ReLU}\left(\text{BN}\left(f_{\text{col}}^{l} \cdot x_{\text{row}}^{h}\right)\right)} + \sqrt{\text{ReLU}\left(\text{BN}\left(x_{\text{col}}^{h} \cdot f_{\text{row}}^{l}\right)\right)} \quad (2)$$

representing $f_l \in \mathbb{R}^{p\times p\times d}$. Matrix multiplication captures the cross-modal interactions, ensuring that the features from each modality enhance the unique characteristics of the other. This interaction allows HSI and LiDAR data to complement each other in the row and column directions, capturing complementary information between the two data. This not only enhances the feature representation capability of central pixels but also extends the complementary information of these features to various regions across both spectral and spatial domains, enabling the model to more accurately integrate and utilize the rich information of both data, thereby enhancing overall feature representation and recognition capabilities.

The original data x_h and x_l, as well as the IEFFs f_h and f_l, underwent feature extraction through a three-layer CNN, each with a 3×3 kernel size. All convolution operations were processed with zero-padding and batch normalization, and nonlinear features were introduced using ReLU activation functions. For the HSI x_h and its corresponding IEFF f_h, features are extracted using three CNN layers, each containing 32 filters. This results in feature representations $X_h \in \mathbb{R}^{p\times p\times 32}$ and $F_h \in \mathbb{R}^{p\times p\times 32}$. For the LiDAR data x_l and its corresponding IEFFs f_l, features are extracted using three CNN layers with filters of 2, 8, and 16, respectively. This results in feature representations $X_l \in \mathbb{R}^{p\times p\times 16}$ and $F_l \in \mathbb{R}^{p\times p\times 16}$.

B. Multihead Self-Attention-Graph Convolutional Network

The convolutional features from the original data (X_h, X_l) and the IEFFs (F_h, F_l) contain abundant spectral and spatial information. To fully utilize this information, we introduced GCN to process the pixels. In this process, each pixel in the data cubes is treated as a node in the graph.

The graph structure $\mathcal{G}$ is represented as $\mathcal{G} = \{G_1, G_2, \ldots, G_i, \ldots, G_k\}$, where $i = 1, 2, \ldots, k$. Here, k denotes the total number of pixel samples to be classified. Each G_i consists of $\{Z_1, \ldots, Z_j, \ldots, Z_4\}$, with $j = 1, 2, 3, 4$. Each feature Z_j, obtained from X_h, F_h, X_l, and F_l, is represented as $Z_j = \{H_j, \tilde{A}(j)\}$. Specifically, each extracted feature is first spatially flattened into H_j, where $H_j \in \mathbb{R}^{p^2\times S}$. Then, the spectral similarity of H_j is calculated to construct the adjacency matrix $\tilde{A}_j$, where $\tilde{A}_j \in \mathbb{R}^{p^2\times p^2}$. The adjacency matrix $\tilde{A}_j$ is calculated using the spectral similarity function, represented as

$$\tilde{A}_j(a, b) = \exp\left(-\frac{\left\|H_j(a) - H_j(b)\right\|^2}{S}\right) \quad (3)$$

where S represents the spectral dimension of the current feature H_j, and a and b range form 1 to p^2.

This article proposes a module that combines GCN and the MSA mechanism to handle data with graph structures called the MSA-GCN. The MSA-GCN module effectively explores spatial and spectral information through MSA mechanism and graph convolution operations, achieving efficient representation and learning of graph structure data.

In the MSA-GCN module, we input Z_j into the MSA-GCN. The lth layer is represented as

$$\hat{H}_j^{l} = \text{ReLU}\left(\tilde{A}_j \cdot H_j^{l} \cdot W^{l}\right), \quad l = 0, 1 \quad (4)$$

where W^l is the trainable matrix of the lth layer. The resulting feature representation $\hat{H}_j^{l}$ has dimensions $\mathbb{R}^{p^2\times S}$.

Graph convolutional features $\hat{H}_j^{l}$ mainly explore spatial information. We apply the MSA mechanism along the spectral dimension to better learn global spectral dependencies while retaining local spatial information. Specifically, the input feature vector $\hat{H}_j^{l}$ is first transformed into queries, keys, and values (i.e., Q^l, K^l, and V^l). The Q^l, K^l, and V^l matrices are then mapped to h latent spaces, and the attention values are,

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finally, calculated. The calculation process of global spectral dependencies based on MSA is given as follows:

$$H_j^{(l+1)} = \mathrm{MSA}\left(Q^l, K^l, V^l\right) = \mathrm{Softmax}\left(\frac{Q^l \cdot \left(K^l\right)^T}{\sqrt{h}}\right) \cdot V^l. \quad (5)$$

Ultimately, the output result H_j^2 is represented as $H_j^2 \in \mathbb{R}^{p^2 \times S}$.

GCN can effectively utilize the graph structure to model spatial relationships among the data, thereby capturing and learning spatial information. The MSA mechanism can learn global dependencies in the spectral dimension by processing spectral information in parallel with multihead. This mechanism focuses on different spectral features in the data, thereby effectively capturing the correlation and complex nonlinear relationships between spectral features. By combining GCN with MSA, the model can simultaneously consider spatial and spectral information. This facilitates the effective transmission and fusion of spectral and spatial information, allowing the model to understand the structure and characteristics of the data more accurately. Consequently, the model's ability to model and analyze complex data is improved.

C. Multimodal Gated Fusion Module

To fully leverage the advantages of HSI and LiDAR data, we designed an MGFM to integrate features from different sources. The core of the fusion module is a gated unit, which effectively combines the features from HSI and LiDAR data, enhancing the model's feature representation capability.

We have four sets of features: F_0^2, F_1^2, F_2^2, and F_3^2. Here, F_0^2 and F_1^2 represent the outputs obtained from the convolutional features and the IEFFs of HSI, respectively, with dimensions $\mathbb{R}^{p^2 \times 32}$. F_2^2 and F_3^2 are obtained from the convolutional features and the IEFFs of LiDAR data, respectively, with dimensions $\mathbb{R}^{p^2 \times 16}$. To further fuse these features, F_2^2 and F_3^2 are concatenated along the spectral dimension, resulting in a new feature F_{Lidar} with dimensions $\mathbb{R}^{p^2 \times 32}$.

Next, the three input features F_0^2, F_{Lidar}, and F_1^2 are concatenated along the spectral dimension, forming a tensor with dimensions $\mathbb{R}^{p^2 \times 96}$. Using dot product and broadcast mechanisms, we calculate the gated scores. Specifically, we multiply the concatenated feature tensor by the gating weight matrix and add a bias term to obtain the initial gated score matrix M. Then, M is reshaped to $\mathbb{R}^{p^2 \times 32 \times 3}$, assigning a gated score to each feature. Applying the softmax function along the spectral dimension yields the gating weights for each feature. The input features are then weighted and summed according to the gating weights to obtain the fused feature output. Specifically, the output result Output is

$$\begin{aligned}\mathrm{Output} = {} & M[\ldots, 0] \times F_0^2 \\ & + M[\ldots, 1] \times F_{\mathrm{Lidar}}^2 \\ & + M[\ldots, 2] \times F_1^2. \end{aligned} \quad (6)$$

The fused feature Output combines the advantages of HSI and LiDAR data, enhancing the model's feature representation capability for multimodal data. This fusion module enables effective information exchange between different features, fully exploiting the complementary advantages of different data sources, thereby improving the model's overall feature representation and recognition capability.

Finally, a fully connected layer with a softmax classifier is applied to the output Output to obtain the probability distribution $\hat{y}$ for each class. To evaluate the classification performance, we use cross-entropy as the loss function, represented as

$$\mathrm{Loss} = -\frac{1}{N}\sum_{i}^{N}\sum_{\mathrm{cl}=1}^{\mathrm{CL}} y_{i,\mathrm{cl}} \log \hat{y}_{i,\mathrm{cl}} \quad (7)$$

where N is the total number of samples, CL is the total number of classes, $y_{i,\mathrm{cl}}$ is the true label of sample i belonging to class cl, and $\hat{y}_{i,\mathrm{cl}}$ is the predicted probability of sample i belonging to class cl.

TABLE I
NUMBER OF SAMPLES IN EACH CLASS OF THE TRENTO DATASET

Class		Number of Samples	
No.	Name	Train	Test
1	Apple Trees	129	3905
2	Buildings	125	2778
3	Ground	105	374
4	Woods	154	8969
5	Vineyard	184	10317
6	Roads	122	3052
-	Total	819	29395

III. EXPERIMENTS

To validate the effectiveness and rationality of the model, we conducted experiments on three publicly available datasets (Trento, MUUFL, and Houston2013) and provided a detailed discussion and analysis of the experimental results. First, we introduce detailed information about these three datasets used in the experiments. Next, we performed ablation experiments on the proposed modules within the framework to illustrate the contribution and unique role of each module in the overall framework. In addition, we compared the performance differences between multimodal and unimodal classifications, demonstrating the advantages of the multimodal approach in handling complex data. Finally, we conducted multiple comparative experiments, thoroughly comparing the proposed method with several existing advanced methods. By presenting detailed experimental results, we demonstrate the effectiveness of the proposed method, particularly in terms of accuracy and robustness, which significantly outperforms existing methods.

A. Dataset Description

We conducted experiments on three publicly available datasets. Specifically, the Trento dataset contains HSI with 63 spectral bands and LiDAR data, both with a spatial resolution of 600 × 166 pixels. Table I provides detailed information on the class names and the number of training and testing samples for the Trento dataset. The MUUFL dataset contains HSI with 64 spectral bands and LiDAR data, both with a spatial resolution of 325 × 220 pixels. Table II provides

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TABLE II
NUMBER OF SAMPLES IN EACH CLASS OF THE MUUFL DATASET

Class		Number of Samples	
No.	Name	Train	Test
1	Trees	150	23096
2	Mostly grass	150	4120
3	Mixed ground surface	150	6732
4	Dirt and sand	150	1676
5	Road	150	6537
6	Water	150	316
7	Building Shadow	150	2083
8	Building	150	6090
9	Sidewalk	150	1235
10	Yellow curb	150	33
11	Cloth panels	150	119
-	Total	1650	52037

TABLE III
NUMBER OF SAMPLES IN EACH CLASS OF THE HOUSTON2013 DATASET

Class		Number of Samples	
No.	Name	Train	Test
1	Health grass	198	1053
2	Stressed grass	190	1064
3	Synthetic grass	192	505
4	Trees	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking lot 1	192	1041
13	Parking lot 2	184	285
14	Tennis court	181	247
15	Running track	187	473
-	Total	2832	12197

detailed information on the class names and the number of training and testing samples for the MUUFL dataset. The Houston2013 dataset contains HSI with 144 spectral bands and LiDAR data, both with a spatial resolution of 349 × 1905 pixels. Table III provides detailed information on the class names and the number of training and testing samples for the Houston2013 dataset.

B. Parameter Setting Details and Performance Analysis

We used the PyTorch framework for model training, employing the Adam optimizer and the cross-entropy loss function. The training was conducted for 150 epochs on the three publicly available datasets with an initial learning rate of 0.001.

1) Impact of Patch Size on Classification Performance: We conducted a detailed analysis of the impact of input patch size on model performance, comparing classification results on the Trento, MUUFL, and Houston2013 datasets. Fig. 2 illustrates the classification performance with different input patch sizes (9 × 9, 11 × 11, 13 × 13, 15 × 15, 17 × 17, and 19 × 19) across the three datasets. The study revealed that the model performs the best with a patch size of 15 × 15, reaching peak overall accuracy (OA) on all three datasets. Increasing the patch size significantly enhances the model's ability to perceive local image features, thereby improving classification accuracy. However, further increasing the patch size, despite capturing more feature information, may introduce additional confusion, affecting the final classification performance. Although there is a slight performance decline with huge patches, the classification performance is still significantly better than that of smaller patches. This improvement can be attributed to the MSA-GCN module in the model, which effectively mitigates the impact of confusing features on useful information. This is discussed further in the ablation experiments section.

TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT COMBINATIONS OF IEFF, MSA-GCN, AND MGFM ON TRENTO, MUUFL, AND HOUSTON2013 DATASETS

Module			Metrics	Datasets		
IEFF	MSA-GCN	MGFM		Trento	MUUFL	Houston
✓	✗	✓	OA(%)	98.08	88.79	91.52
			AA(%)	**97.79**	91.24	92.51
			Kappa	0.9743	0.8535	0.9079
✗	✓	✓	OA(%)	98.31	90.61	90.08
			AA(%)	97.01	92.24	90.79
			Kappa	0.9774	0.8767	0.8922
✓	✓	✗	OA(%)	97.90	88.73	91.47
			AA(%)	97.23	91.62	92.34
			Kappa	0.9719	0.8531	0.9047
✓	✓	✓	OA(%)	**99.21**	**91.56**	**93.20**
			AA(%)	97.02	**92.92**	**93.72**
			Kappa	**0.9894**	**0.8887**	**0.9261**

2) Learned Feature Visualizations by t-SNE: Fig. 3(a)–(c), respectively, shows the t-distributed stochastic neighbor embedding (t-SNE) visualization results of different datasets, revealing the variations in dataset complexity. Fig. 3(a) displays the distribution of various classes in the Trento dataset. In Fig. 3(a), the boundaries between different classes are obvious, with almost no overlap, indicating a high degree of distinguishability in this dataset. Each class forms a distinct cluster, demonstrating good intraclass cohesion and interclass separation. In contrast, Fig. 3(b) presents the class distribution of the MUUFL dataset. Compared to Fig. 3(a), the boundaries between different classes in the MUUFL dataset begin to blur, with a certain degree of overlap and mixing. Fig. 3(c) shows the class distribution in the Houston2013 dataset. Although most classes still exhibit high distinguishability, significant overlap occurs between some classes in certain regions, making the boundaries between them unclear. In summary, the visualizations clearly show that each class in different datasets forms distinct clusters, but their distinguishability and separation vary. As the number of classes increases, distinguishing between classes becomes more challenging.

C. Ablation Studies

1) Ablation Analysis of Different Modal Data Inputs: As shown in Fig. 4, through comparative experiments on the Trento, MUUFL, and Houston2013 datasets using unimodal (HSI or LiDAR) and multimodal (HSI + LiDAR) approaches,

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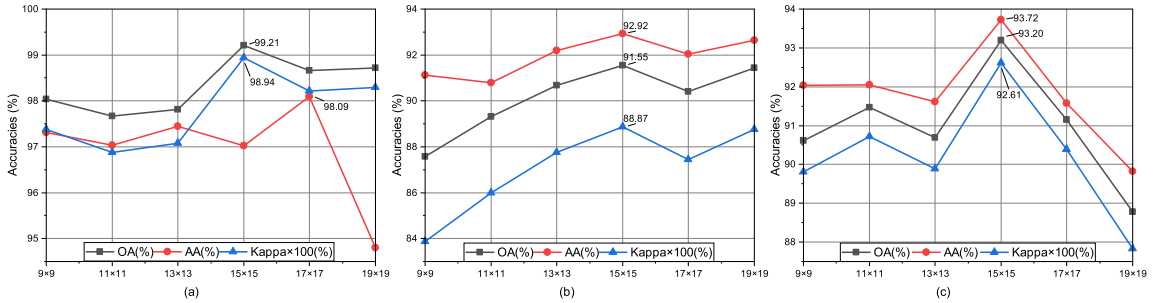


Fig. 2. Classification performance analysis for the proposed method with different patch sizes. (a) Trento. (b) MUUFL. (c) Houston2013.

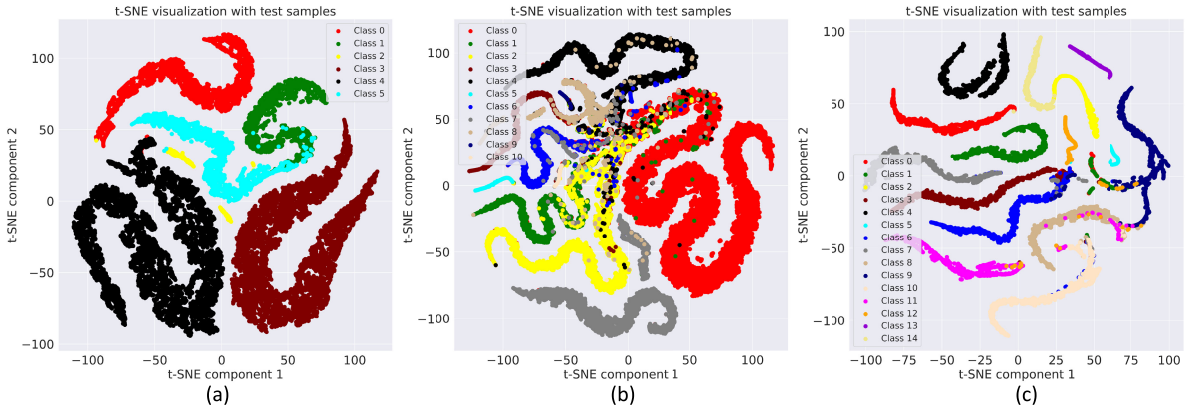


Fig. 3. Visualization of t-SNE results on different datasets. (a) Trento. (b) MUUFL. (c) Houston2013.

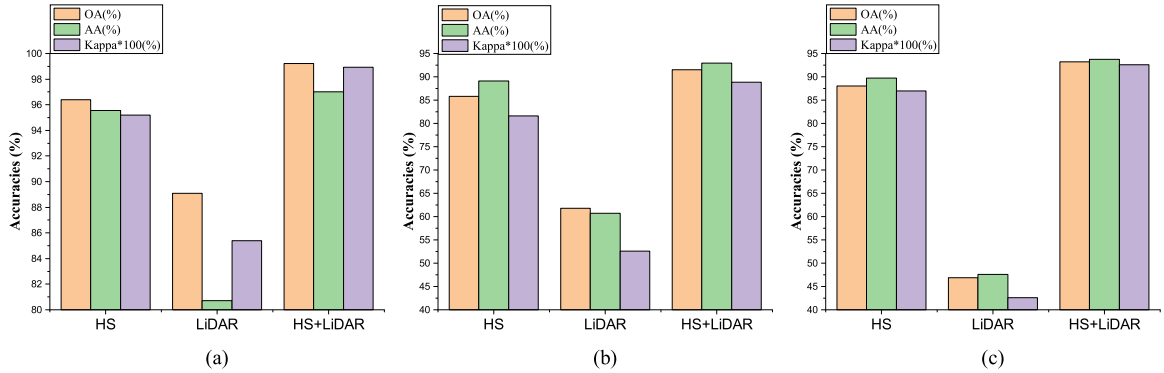


Fig. 4. Ablation studies results in terms of OA(%), AA(%), and Kappa(%) about the data source usage (i.e., only HSI, only LiDAR, and HSI + LiDAR). (a) Trento. (b) MUUFL. (c) Houston2013.

we can clearly observe their significant differences in classification performance, further demonstrating the advantages of multimodal fusion. In the Trento dataset, the OA of the HSI + LiDAR approach exceeded 99%, which is approximately 3% and 10% higher than using HSI and LiDAR alone, respectively. In addition, the average accuracy (AA) and Kappa also showed substantial improvements, reaching around 97% and 0.99, respectively. This indicates that multimodal fusion has significant advantages in recognition accuracy and consistency. In the MUUFL dataset, multimodal fusion showed substantial performance improvement, with HSI + LiDAR achieving an OA of approximately 91.5%, an increase of about 5% and 30% compared to HSI and LiDAR alone, respectively. The AA and Kappa are also significantly higher compared to the unimodal levels, reaching about 93% and 0.89, respectively. In the Houston2013 dataset, multimodal fusion achieved an OA exceeding 93%, improving by approximately 5% and 47% compared to HSI and LiDAR alone, respectively. The AA and Kappa reached approximately 93.7% and 0.926, respectively.

These data clearly demonstrate the powerful advantages of multimodal fusion by effectively integrating complementary information from HSI and LiDAR data, significantly enhancing classification accuracy and model robustness. HSI provides rich spectral information, while LiDAR data offers precise height and structural information. The combination of both modalities allows for a more comprehensive description of terrain features, thereby achieving higher classification accuracy.

2) Ablation Analysis of Different Modules: To fully demonstrate the specific impact of each proposed module on

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TABLE V
CLASSIFICATION PERFORMANCE OF MSA AND GCN COMPONENTS ON TRENTO, MUUFL, AND HOUSTON2013 DATASETS

MSA-GCN		Metrics	Datasets		
GCN	MSA		Trento	MUUFL	Houston
✓	✗	OA(%)	97.98	89.60	90.92
		AA(%)	**97.68**	91.62	92.15
		Kappa	0.9730	0.8638	0.9014
✗	✓	OA(%)	97.94	88.48	91.85
		AA(%)	96.28	90.26	92.69
		Kappa	0.9725	0.8496	0.9115
✓	✓	OA(%)	**99.21**	**91.56**	**93.20**
		AA(%)	97.02	**92.92**	**93.72**
		Kappa	**0.9894**	**0.8887**	**0.9261**

classification performance, we conducted ablation experiments across three datasets, as shown in Table IV. The experiments compared the performance of the IEFF module, MSA-GCN, and MGFM in different combinations. We analyzed the effects of omitting each module individually on the overall model performance, showcasing the critical roles of these modules in joint classification.

The IEFF module enhances feature representation by extracting and integrating row and column features from HSI and LiDAR data, significantly boosting the feature expression capability of central pixels. When IEFF is absent (using MSA-GCN and MGFM), on the Trento dataset, the OA is 98.31%, AA is 97.01%, and Kappa is 0.9774. Compared to the whole module combination, OA decreased by 0.90%, and Kappa decreased by 0.012. In the MUUFL dataset, without IEFF, OA and Kappa are 90.61% and 0.8767, respectively, while, with the whole module combination, they are 91.55% and 0.8887, showing decreases of 0.94% and 0.012, respectively. In the Houston2013 dataset, without IEFF, OA, AA, and Kappa are 90.08%, 90.79%, and 0.8922, respectively, whereas, with the whole module combination, they are 93.20%, 93.72%, and 0.9261, indicating decreases of 3.12%, 2.93%, and 0.0339, respectively. These results underscore the critical role of the IEFF module in enhancing feature representation and integrating complementary information from HSI and LiDAR data.

The MSA-GCN module combines GCN with MSA mechanisms to perform feature learning across spatial and spectral dimensions. On the Trento dataset, without MSA-GCN, the OA is 98.08%, and the Kappa is 0.9743. Compared to the whole module combination, this represents decreases of 1.13% and 0.0151 in OA and Kappa, respectively. In the MUUFL dataset, without MSA-GCN, OA, AA, and Kappa are 88.79%, 91.24%, and 0.8535, respectively, while, with the whole module combination, they are 91.55%, 92.92%, and 0.8887, showing decreases of 2.76%, 1.68%, and 0.0352. In the Houston2013 dataset, without MSA-GCN, OA, AA, and Kappa are 91.52%, 92.51%, and 0.9079, respectively, while, with the whole module combination, they are 93.20%, 93.72%, and 0.9261, indicating decreases of 1.68%, 1.26%, and 0.0182. These data indicate that the MSA-GCN module significantly contributes to capturing spatial information and extracting features. The GCN effectively captures spatial relationships between pixels. At the same time, the MSA mechanism learns global dependencies in the spectral dimension, enabling the module to integrate and utilize spatial and spectral information effectively.

The MGFM integrates features from HSI and LiDAR data using gated units. Without the MGFM (using IEFF and MSA-GCN), the OA and Kappa on the Trento dataset are 97.90% and 0.9719, respectively. Compared to the whole module combination, this represents decreases of 1.31% and 0.0175 in OA and Kappa, respectively. In the MUUFL dataset, without MGFM, OA, AA, and Kappa are 88.73%, 91.62%, and 0.8531, respectively, while, with the whole module combination, they are 91.55%, 92.92%, and 0.8887, showing decreases of 2.82%, 1.3%, and 0.0356. In the Houston2013 dataset, without MGFM, OA, AA, and Kappa are 91.47%, 92.34%, and 0.9047, respectively, while, with the whole module combination, they are 93.20%, 93.72%, and 0.9261, indicating decreases of 1.73%, 1.38%, and 0.0214. These results demonstrate that the MGFM module effectively facilitates information exchange between different features, harnessing the complementary advantages of diverse data sources.

Overall, the proposed joint classification method effectively harnesses the complementary information of HSI and LiDAR data through the integration of various modules, significantly enhancing classification performance. Particularly notable is the improvement in OA across all datasets, highlighting the effectiveness of the approach. This indicates that the combination of IEFF, MSA-GCN, and MGFM allows for a more accurate understanding and utilization of the structure and features of complex data. Consequently, it enhances overall feature representation and recognition capabilities.

3) Ablation Analysis of MSA-GCN: We have analyzed the importance of GCN and MSA combinations in detail in the MSA-GCN module, as shown in Table V. On the one hand, in the Trento dataset, without the MSA component, the OA decreased to 97.98% from 99.21% with the whole module combination, representing a decrease of 1.23%. The Kappa also decreased from 0.9894 to 0.9730, a decrease of 0.0164. In the MUUFL dataset, the absence of the MSA module resulted in an OA of 89.60% and AA of 91.62% compared to 91.55% and 92.92% with the whole module combination, showing decreases of 1.95% and 1.30%, respectively. The Kappa decreased from 0.8887 to 0.8638, a decrease of 0.0249. In the Houston2013 dataset, without the MSA module, OA and AA were 90.92% and 92.15%, respectively, compared to 93.20% and 93.72% with the whole module combination, indicating decreases of 2.28% and 1.57%, respectively. The Kappa decreased from 0.9261 to 0.9014, a decrease of 0.0247. These results demonstrate that the MSA module plays a crucial role in enhancing the model's global dependence on spectral information and its ability to capture features. The significant performance drops across different datasets without the MSA module underscore its critical role in improving classification accuracy and stability.

On the other hand, in the Trento dataset, the absence of the GCN module resulted in an OA of 97.94% and AA of 96.28% compared to 99.21% and 97.02% with the whole module combination, showing decreases of 1.27% and 0.74%,

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TABLE VI
COMPARISON OF THE CLASSIFICATION ACCURACIES USING THE TRENTO DATASET

No.	Class Name	Performance							
		AMKSC	CALC	GAMF	GLT-Net	M2F	MS2CANet	S^2ENet	Proposed
1	Apple Trees	99.45	97.26	96.49	99.59	**99.77**	99.23	99.33	99.62
2	Buildings	94.11	**99.89**	99.53	99.21	99.32	98.88	99.75	98.81
3	Ground	0.00	80.75	97.86	**99.20**	98.4	92.51	97.33	88.24
4	Woods	99.91	**100.00**	99.99	**100.00**	98.39	**100.00**	**100.00**	99.99
5	Vineyard	57.20	99.90	99.96	**100.00**	99.93	99.90	**100.00**	**100.00**
6	Roads	9.07	90.33	90.53	92.04	89.55	92.10	90.43	**95.45**
	OA(%)	73.32	98.34	98.46	99.03	98.29	98.84	98.86	**99.21**
	AA(%)	59.96	94.69	97.39	**98.34**	97.56	97.10	97.81	97.02
	Kappa	0.6591	0.9778	0.9794	0.9871	97.71	0.9845	0.9847	**0.9894**

TABLE VII
COMPARISON OF THE CLASSIFICATION ACCURACIES USING THE MUUFL DATASET

No.	Class Name	Performance							
		AMKSC	CALC	GAMF	GLT-Net	M2F	MS2CANet	S^2ENet	Proposed
1	Trees	55.05	**97.71**	90.86	93.07	90.92	92.73	94.29	92.69
2	Mostly grass	72.86	83.20	82.45	86.87	85.32	85.61	66.50	**87.45**
3	Mixed ground surface	29.42	76.31	78.98	82.83	**86.96**	83.60	80.39	86.66
4	Dirt and sand	73.88	93.74	94.45	91.89	96.48	**97.26**	94.27	96.24
5	Road	55.14	88.74	90.48	91.75	90.21	88.85	**92.95**	91.17
6	Water	0.00	**100.00**	99.68	99.68	95.25	**100.00**	99.37	99.68
7	Building Shadow	73.09	90.49	90.16	94.72	92.80	**95.39**	91.45	94.48
8	Building	21.75	94.61	94.50	94.94	91.33	**97.83**	96.39	94.71
9	Sidewalk	25.05	63.48	83.56	85.67	74.49	84.86	**85.91**	82.91
10	Yellow curb	0.00	87.88	**100.00**	93.94	93.94	93.94	90.91	96.97
11	Cloth panels	**99.63**	99.16	96.64	99.16	98.32	99.16	99.16	99.16
	OA(%)	49.50	91.09	89.03	91.22	89.83	91.22	90.10	**91.56**
	AA(%)	45.99	88.67	91.07	92.23	90.55	92.66	90.14	**92.92**
	Kappa	0.3978	0.8806	0.8562	0.8844	0.8668	0.8856	0.8686	**0.8887**

TABLE VIII
COMPARISON OF THE CLASSIFICATION ACCURACIES USING THE HOUSTON2013 DATASET

No.	Class Name	Performance							
		AMKSC	CALC	GAMF	GLT-Net	M2F	MS2CANet	S^2ENet	Proposed
1	Health grass	14.23	82.53	81.86	83.00	**88.51**	83.00	84.24	83.00
2	Stressed grass	72.57	84.40	83.18	84.77	84.49	85.15	**100.00**	85.15
3	Synthetic grass	0.00	90.50	**100.00**	92.08	94.46	96.24	**100.00**	97.82
4	Trees	77.81	91.10	99.72	93.28	98.48	97.92	**100.00**	92.61
5	Soil	86.47	**100.00**	**100.00**	**100.00**	98.39	**100.00**	**100.00**	**100.00**
6	Water	0.00	95.10	94.41	**100.00**	95.80	**100.00**	99.30	96.50
7	Residential	37.62	88.06	86.66	85.26	94.03	**95.43**	88.99	92.63
8	Commercial	21.14	75.78	72.74	93.64	91.17	86.80	**95.54**	91.83
9	Road	24.12	88.01	81.11	91.22	91.50	85.74	89.14	**94.43**
10	Highway	42.05	68.24	58.69	64.09	70.56	83.69	77.61	**95.17**
11	Railway	27.04	**99.72**	77.70	82.54	89.94	95.07	96.02	97.34
12	Parking lot 1	59.77	96.54	95.29	94.52	95.58	**97.21**	85.11	95.10
13	Parking lot 2	0.00	91.23	82.11	**91.58**	90.18	86.67	82.46	84.56
14	Tennis court	0.00	**100.00**	**100.00**	99.60	89.88	90.69	99.60	**100.00**
15	Running track	**100.00**	99.58	96.83	**100.00**	**100.00**	**100.00**	**100.00**	99.58
	OA(%)	42.71	88.49	85.33	88.46	90.90	91.57	92.40	**93.20**
	AA(%)	37.52	90.05	87.35	90.37	91.53	92.24	93.20	**93.72**
	Kappa	0.3811	0.8754	0.8409	0.8752	0.9014	0.9092	0.9175	**0.9261**

respectively. The Kappa decreased from 0.9894 to 0.9725, a decrease of 0.0169. In the MUUFL dataset, without the GCN module, OA was 88.48% and AA was 90.26% compared to 91.56% and 92.92% with the whole module combination, showing decreases of 3.08% and 2.66%, respectively. The Kappa decreased from 0.8887 to 0.8496, a decrease of 0.0391. In the Houston2013 dataset, the absence of the GCN module resulted in an OA of 91.85% and AA of 92.69% compared to 93.20% and 93.72% with the whole module combination, showing decreases of 1.35% and 1.03%, respectively. The

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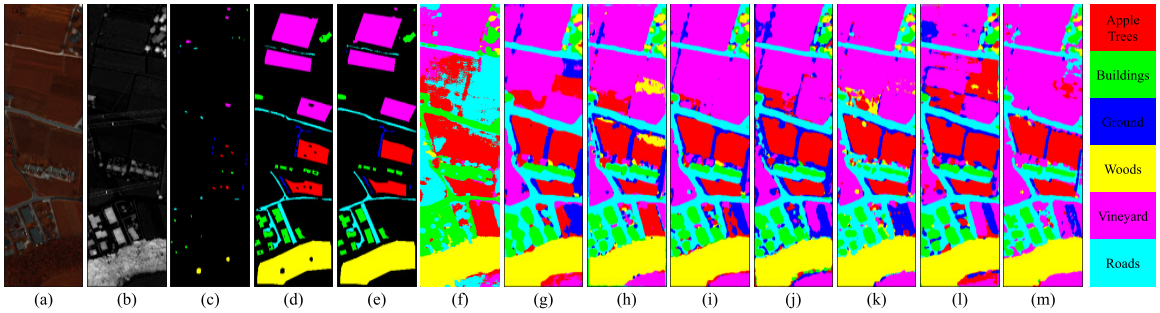


Fig. 5. Classification maps for the Trento data obtained with different methods. (a) False-color HSI (using bands 40, 20, and 10 as R, G, and B, respectively). (b) Gray image for LiDAR. (c) Training map. (d) Testing map. (e) Ground-truth map. (f) AMKSC. (g) CALC. (h) GAMF. (i) GLT-Net. (j) M2F. (k) MS2CANet. (l) S^2ENet. (m) Proposed.

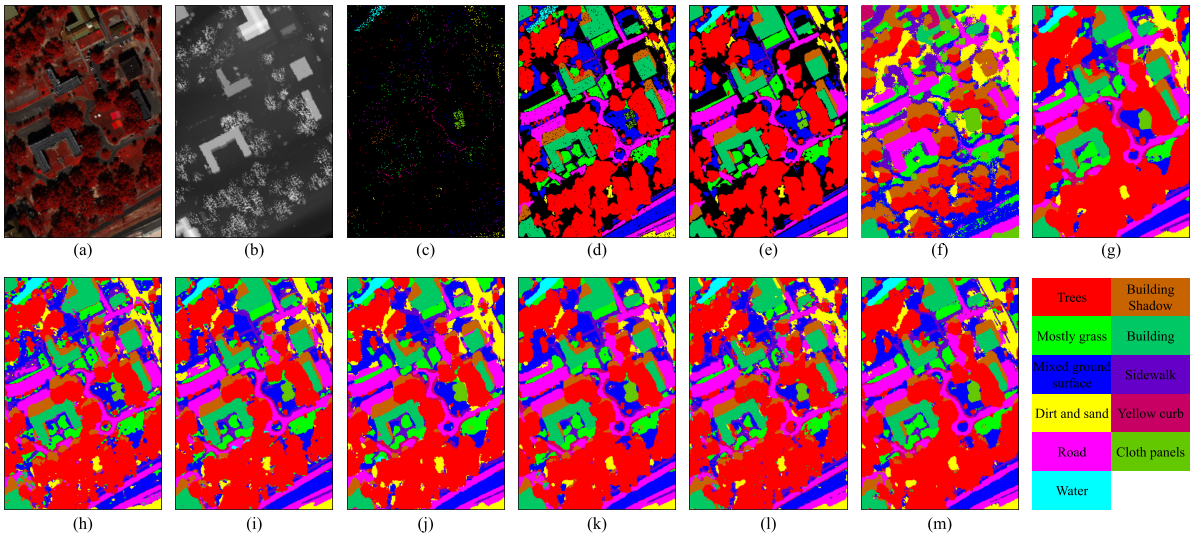


Fig. 6. Classification maps for the MUUFL data obtained with different methods. (a) False-color HSI (using bands 40, 20, and 10 as R, G, and B, respectively). (b) Gray image for LiDAR. (c) Training map. (d) Testing map. (e) Ground-truth map. (f) AMKSC. (g) CALC. (h) GAMF. (i) GLT-Net. (j) M2F. (k) MS2CANet. (l) S^2ENet. (m) Proposed.

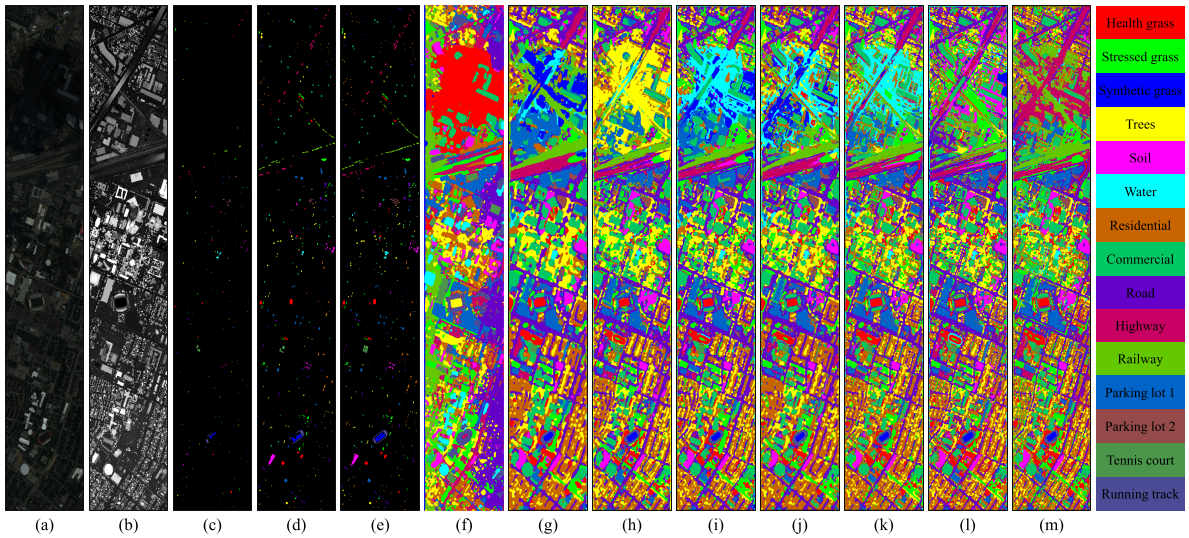


Fig. 7. Classification maps for the Houston2013 data obtained with different methods. (a) False-color HSI (using bands 40, 20, and 10 as R, G, and B, respectively). (b) Gray image for LiDAR. (c) Training map. (d) Testing map. (e) Ground-truth map. (f) AMKSC. (g) CALC. (h) GAMF. (i) GLT-Net. (j) M2F. (k) MS2CANet. (l) S^2ENet. (m) Proposed.

Kappa decreased from 0.9261 to 0.9115, a decrease of 0.0146. These results demonstrate that the GCN module plays a crucial role in extracting and integrating spatial relationship features. The significant performance drops across different datasets without the GCN module underscore its necessity in capturing spatial information and improving classification performance.

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In summary, the MSA module and the GCN module play crucial roles in the joint classification of HSI and LiDAR data. The MSA module captures global dependencies of spectral information, integrates comprehensive contextual information, and identifies complex spectral features, significantly enhancing model accuracy and consistency. Meanwhile, the GCN module aggregates local spatial information, effectively extracts and integrates spatial relational features, and strengthens the model's ability to model complex data. The MSA-GCN module leverages and integrates spatial and spectral information comprehensively, thereby significantly improving classification accuracy and model stability, highlighting the indispensable role of the MSA-GCN module in enhancing overall model performance.

D. Comparative Experiments and Analysis

We conducted a series of comparative experiments to validate our proposed network for joint classification of HSI and LiDAR data. Tables VI–VIII present these experiments on three public datasets: Trento, MUUFL, and Houston2013. We compared our method with several state-of-the-art multimodal data fusion methods, including AMKSC [60], CALC [34], GAMF [58], GLT-Net [50], M2F [37], MS2CANet [27], and spatial–spectral cross-modal enhancement network (S^2ENet) [33]. These methods employ various strategies such as GCN, generative adversarial networks (GANs), and Transformer architectures, representing different approaches to multimodal data fusion and classification models.

1) Comparative of Different Classification Methods: In the Trento dataset, we conducted detailed comparisons of different methods' classification performance, as shown in Table VI. The results demonstrate significant advantages of our proposed method in OA and Kappa. Specifically, our method achieved OA and Kappa of 99.21% and 0.9894, respectively. In contrast, other methods such as CALC, GAMF, GLT-Net, M2F, and MS2CANet achieved OA values of 98.34%, 98.46%, 99.03%, 98.29%, and 98.84%, respectively. While some methods approached or slightly exceeded our method in AA, our method clearly outperformed them in OA and Kappa. This indicates that while other methods excel in certain classes, they fall slightly short in overall performance consistency across diverse land cover types. Specifically, our method exceled in classifying the "road" and "vineyard" categories, with AA values reaching 95.45% and 100%, respectively. This highlights our method's robustness and accuracy in handling complex and easily confused land cover categories. This advantage is attributed to the efficiency of our model in fusing multimodal data (HSI and LiDAR data), allowing better capture and utilization of spatial and spectral features of land cover, thereby improving classification accuracy and consistency. In addition, the MSA module effectively captures long-distance dependencies, while the GCN module excels in handling spatial structural information, enhancing the overall performance of our model.

Our proposed method also demonstrated outstanding classification performance in the MUUFL dataset. As shown in Table VII, our method achieved OA, AA, and Kappa of 91.56%, 92.92%, and 0.8887, respectively. In comparison, other methods such as CALC, GAMF, GLT-Net, M2F, and MS2CANet achieved OA values of 91.09%, 89.03%, 91.22%, 89.83%, and 91.22%, respectively. This significant performance improvement is attributed to the more comprehensive information representation facilitated by the fusion of multimodal data.

In the Houston2013 dataset, our proposed method also demonstrated excellent performance with OA, AA, and Kappa of 93.20%, 93.72%, and 0.9261, respectively. In contrast, other methods such as CALC, GAMF, GLT-Net, M2F, and MS2CANet achieved OA values of 88.49%, 85.33%, 88.46%, 90.90%, and 91.57%, respectively, as shown in Table VIII. From Table VIII, it is evident that our proposed method exceled across multiple categories, particularly achieving high accuracy in the "Highway" and "Road" categories with accuracies of 95.17% and 94.43%, respectively, which significantly outperformed other methods. This substantial advantage exemplifies the power of our proposed multimodal data fusion approach in improving the classification accuracy of HSI and LiDAR data. By combining HSI and LiDAR data, our model effectively utilizes both spectral and spatial information to more accurately capture and distinguish features of different land cover classes. Furthermore, the MSA module enhances the model's ability to understand complex relationships between different classes, while the GCN module efficiently processes spatial features, further enhancing the model's classification performance. The combination of these techniques allows our proposed method to maintain high accuracy and consistency in classifying the complex and varied terrain of the Houston2013 dataset.

2) Visual Comparison: Figs. 5–7 sequentially depict on the Trento, MUUFL, and Houston2013 datasets the pseudo-color composite images of HSI, gray-scale images of LiDAR-based DSM, training set images, test set images, ground-truth maps, and visual classification results of different methods. Figs. 5–7 also show the colors corresponding to each class.

As shown in Fig. 5, when analyzing the classification maps of the Trento dataset, our method demonstrates smoother classification results with fewer isolated points compared to other methods. This is attributed to our model's capability of information fusion and the application of the MSA mechanism. By integrating GCNs and MSA, our model effectively captures and synthesizes features from different modalities, thereby reducing noise and errors in the classification process. Specifically, the classification results closely match the ground-truth map, highlighting the superior performance of our method in handling complex and noisy data environments.

As depicted in Fig. 6, on the MUUFL dataset, our method shows better capability in distinguishing mixed land parcels. Through the IEFF module, MSA-GCN, our approach can delineate individual categories within the entire area more clearly and accurately, resulting in reduced noise. This advantage is particularly crucial when dealing with HSI, which typically contains extensive spectral information that can be effectively utilized by our MGFM to enhance classification accuracy and reliability.

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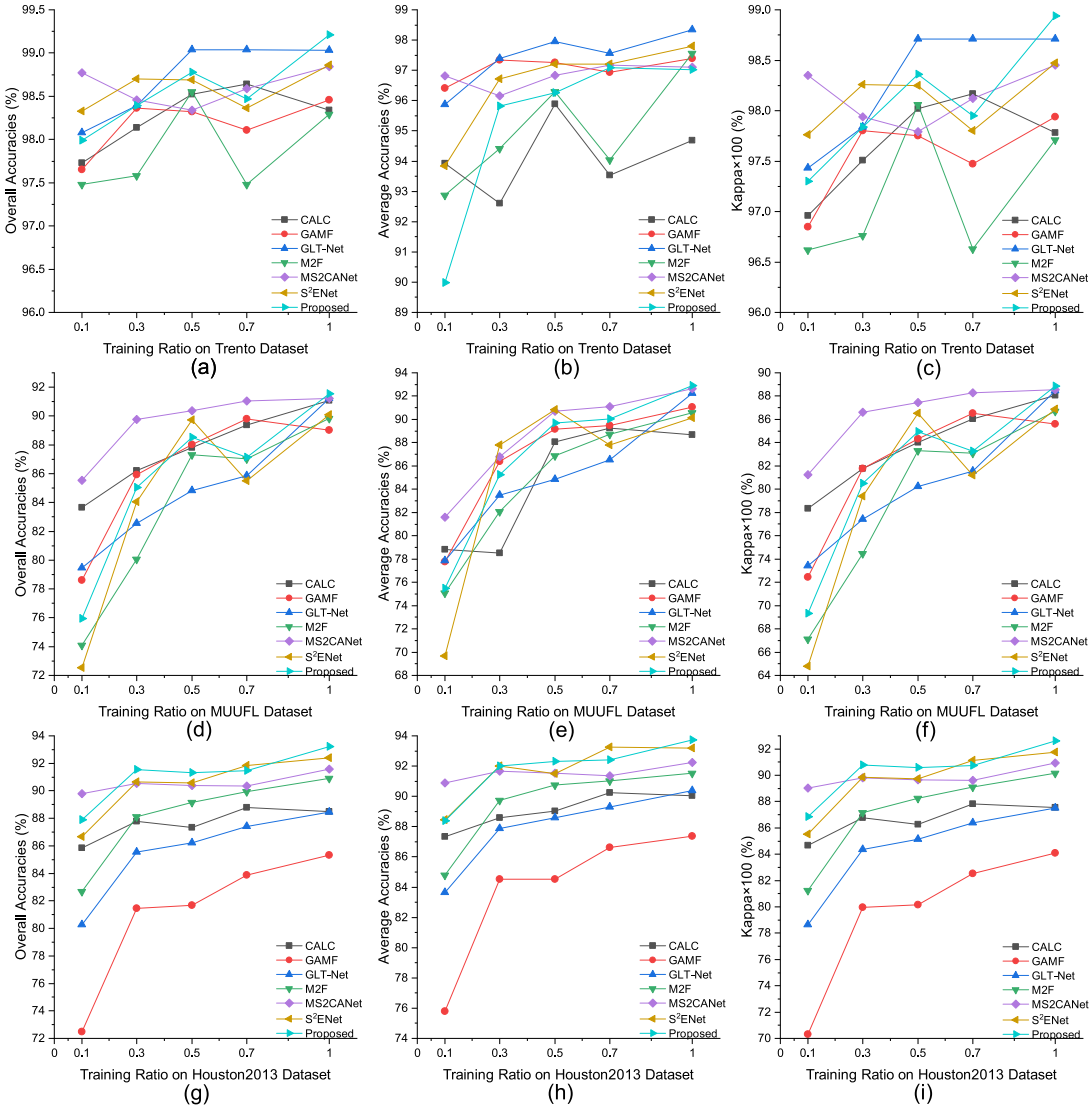


Fig. 8. Impact of different training ratios on classification performance in terms of OA(%), AA(%), and Kappa × 100 (%) across various methods applied to three datasets. (a)–(c) OA, AA, and Kappa on the Trento dataset, respectively. (d)–(f) OA, AA, and Kappa on the MUUFL dataset, respectively. (g)–(i) OA, AA, and Kappa on the Houston2013 dataset, respectively.

As shown in Fig. 7, on the Houston2013 dataset, where some areas are obscured by clouds or fog, our model relies heavily on LiDAR data to supplement the elevation information of HSI data. Our model leverages the complementary information from multimodal data more effectively, especially in cloud-obscured areas, by utilizing terrain information provided by LiDAR data to compensate for the shortcomings of HSI. Visual classification maps indicate that while other models exhibit blurred classification results for Highway and Road categories, our proposed method accurately displays the textures of these areas and distinguishes them more precisely from surrounding features.

3) Performance Comparison for Different Training Ratio: It can be observed from Fig. 8 that despite some methods (such as GLT-Net and MS2CANet) approaching or slightly surpassing our method at specific training ratios, our approach exhibits significant advantages in OA and Kappa when the training ratio is 1. Although the results on the Trento and MUUFL datasets appear somewhat scattered, I attribute this to the datasets having fewer training classes, leading to larger performance variations among methods. Nevertheless, our method performs exceptionally well under these circumstances, showcasing its strengths in handling small sample sizes and class imbalance issues. On the Houston2013 dataset, our proposed method consistently outperforms other comparison methods at training ratios from 0.3 to 1 [as shown in Fig. 8(g)–(i)], demonstrating its robust capability to maintain high performance even under conditions with limited training data and complex terrain. As the training ratio increases, our method continues to excel, indicating its adaptability and robustness in tackling complex and diverse land classification tasks. From Fig. 8, it is evident that while different methods tend to converge in performance at higher training ratios (0.7 and 1.0), our proposed method significantly outperforms other methods at lower training ratios. This highlights our method's clear advantage in handling small sample size issues,

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enabling more accurate classification with limited training data.

IV. Conclusion

In this article, we propose a new interactive enhanced network based on MSA and GCN for the classification of HSI and LiDAR data. Individual features are extracted from HSI and LiDAR data, and IEFFs are constructed based on the row and column features of the central pixel. Individual features focus on the local properties of a single data source, while IEFFs strengthen the feature representation of the central pixel through matrix operations and integrate complementary information from both data sources. Next, we combine MSA and GCN for feature extraction. Specifically, GCN is used to construct graph structures for the four types of features, modeling pixels as nodes to capture spatial relationships. The MSA mechanism is then applied to learn global dependencies in the spectral dimension, processing spectral information in parallel with multiple heads to focus on different spectral features, thereby better capturing correlations between spectral features. Finally, through a multimodal gated fusion mechanism, weighted summation is performed on multiple spectral–spatial features extracted, which are then combined with the original features to generate a comprehensive feature representation. The excellent classification performance on three mainstream datasets demonstrates the effectiveness of our proposed method.

In the current method, there may be limitations in capturing deep cross-modal associations. Future improvements could focus on introducing adaptive deep fusion mechanisms that learn complex nonlinear relationships between different modal features and dynamically adjust the fusion strategy. In addition, while the combination of MSA and GCN enhances feature representation capabilities, it also increases computational complexity, particularly when handling large-scale datasets or high-resolution images, leading to high computational resource demands. Therefore, future research could aim at optimizing computational efficiency through model compression or acceleration techniques while also exploring more advanced feature fusion methods to better integrate information from multiple data sources.

References

[1] P. Duan, X. Kang, P. Ghamisi, and S. Li, "Hyperspectral remote sensing benchmark database for oil spill detection with an isolation forest-guided unsupervised detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509711.

[2] P. Duan, Z. Xie, X. Kang, and S. Li, "Self-supervised learning-based oil spill detection of hyperspectral images," *Sci. China Technolog. Sci.*, vol. 65, no. 4, pp. 793–801, Apr. 2022.

[3] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS3LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.

[4] L. Gao, X. Sun, X. Sun, L. Zhuang, Q. Du, and B. Zhang, "Hyperspectral anomaly detection based on chessboard topology," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505016.

[5] Y. Zhang et al., "Depthwise separable convolutional autoencoders for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[6] P. Ghamisi et al., "Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 6–39, Mar. 2019.

[7] S. Kahraman and R. Bacher, "A comprehensive review of hyperspectral data fusion with LiDAR and SAR data," *Annu. Rev. Control*, vol. 51, pp. 236–253, Jan. 2021.

[8] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.

[9] A. Kuras, M. Brell, J. Rizzi, and I. Burud, "Hyperspectral and LiDAR data applied to the urban land cover machine learning and neural-network-based classification: A review," *Remote Sens.*, vol. 13, no. 17, p. 3393, Aug. 2021.

[10] S. J. Buckley, T. H. Kurz, J. A. Howell, and D. Schneider, "Terrestrial LiDAR and hyperspectral data fusion products for geological outcrop analysis," *Comput. Geosci.*, vol. 54, pp. 249–258, Mar. 2013.

[11] E. Tusa et al., "Fusion of hyperspectral imaging and LiDAR for forest monitoring," in *Data Handling in Science and Technology*, vol. 32. Amsterdam, The Netherlands: Elsevier, 2019, pp. 281–303.

[12] G. Rivera, R. Porras, R. Florencia, and J. P. Sánchez-Solís, "LiDAR applications in precision agriculture for cultivating crops: A review of recent advances," *Comput. Electron. Agricult.*, vol. 207, Apr. 2023, Art. no. 107737.

[13] Q. Man, P. Dong, and H. Guo, "Pixel-and feature-level fusion of hyperspectral and LiDAR data for urban land-use classification," *Int. J. Remote Sens.*, vol. 36, no. 6, pp. 1618–1644, 2015.

[14] T. Sankey, J. Donager, J. McVay, and J. B. Sankey, "UAV LiDAR and hyperspectral fusion for forest monitoring in the southwestern USA," *Remote Sens. Environ.*, vol. 195, pp. 30–43, Jun. 2017.

[15] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2023.

[16] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509417.

[17] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503914.

[18] L. Gao, Z. Wang, L. Zhuang, H. Yu, B. Zhang, and J. Chanussot, "Using low-rank representation of abundance maps and nonnegative tensor factorization for hyperspectral nonlinear unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504017.

[19] B. Tu, W. He, Q. Li, Y. Peng, and A. Plaza, "A new context-aware framework for defending against adversarial attacks in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505114.

[20] H. Gao, Y. Zhang, Z. Chen, and C. Li, "A multiscale dual-branch feature fusion and attention network for hyperspectral images classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8180–8192, 2021.

[21] M. Zhang, F. Gao, J. Dong, and L. Qi, "Multi-scale feature fusion for hyperspectral and LiDAR data joint classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 2856–2859.

[22] H. Gao, H. Feng, Y. Zhang, S. Xu, and B. Zhang, "AMSSE-Net: Adaptive multiscale spatial–spectral enhancement network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531317.

[23] X. He and Y. Chen, "Transferring CNN ensemble for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 876–880, May 2021.

[24] M. Kanthi, T. H. Sarma, and C. S. Bindu, "A 3D-deep CNN based feature extraction and hyperspectral image classification," in *Proc. IEEE India Geosci. Remote Sens. Symp. (InGARSS)*, Dec. 2020, pp. 229–232.

[25] A. Wang et al., "Collaborative classification of hyperspectral and LiDAR data based on dual-branch convolutional neural network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 2738–2741.

[26] S. K. Roy, A. Deria, D. Hong, M. Ahmad, A. Plaza, and J. Chanussot, "Hyperspectral and LiDAR data classification using joint CNNs and morphological feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530416.

[27] X. Wang, J. Zhu, Y. Feng, and L. Wang, "MS2CANet: Multi-scale spatial–spectral cross-modal attention network for hyperspectral image and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

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[28] R. Hanachi, A. Sellami, I. R. Farah, and M. D. Mura, "Multi spectral–spatial Gabor feature fusion based on end-to-end deep learning for hyperspectral image classification," in *Proc. 12th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2022, pp. 1–6.

[29] Y. Peng, Y. Zhang, B. Tu, C. Zhou, and Q. Li, "Multiview hierarchical network for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1454–1469, 2022.

[30] Y. Liu, Z. Ye, Y. Xi, H. Liu, W. Li, and L. Bai, "Multiscale and multidirection feature extraction network for hyperspectral and LiDAR classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9961–9973, 2024.

[31] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506812.

[32] M. Zhang, W. Li, Y. Zhang, R. Tao, and Q. Du, "Hyperspectral and LiDAR data classification based on structural optimization transmission," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3153–3164, May 2023.

[33] S. Fang, K. Li, and Z. Li, "S^{2}ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[34] T. Lu, K. Ding, W. Fu, S. Li, and A. Guo, "Coupled adversarial learning for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 93, pp. 118–131, May 2023.

[35] H. Yang, H. Yu, K. Zheng, J. Hu, T. Tao, and Q. Zhang, "Hyperspectral image classification based on interactive transformer and CNN with multilevel feature fusion network," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[36] S. Xia, X. Zhang, H. Meng, and L. Jiao, "Ternary modality contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522017.

[37] L. Sun, X. Wang, Y. Zheng, Z. Wu, and L. Fu, "Multiscale 3-D–2-D mixed CNN and lightweight attention-free transformer for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 2100116.

[38] W. Wang et al., "Dual-branch feature fusion network based cross-modal enhanced CNN and transformer for hyperspectral and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[39] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.

[40] T. Song, Z. Zeng, C. Gao, H. Chen, and J. Li, "Joint classification of hyperspectral and LiDAR data using height information guided hierarchical fusion-and-separation network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505315.

[41] M. Feng, F. Gao, J. Fang, and J. Dong, "Hyperspectral and LiDAR data classification based on linear self-attention," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jun. 2021, pp. 2401–2404.

[42] W. Song, Z. Gao, L. Fang, and Y. Zhang, "A new multi-level attention feature fusion method for hyperspectral and LiDAR data joint classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 5978–5981.

[43] Y. Cheng, Y. J. Deng, W. Y. Wang, C. F. Long, and X. H. Zhu, "Locality preserved MLP-mixer for hyperspectral image classification," in *Proc. 13th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Oct./Nov. 2023, pp. 1–5.

[44] Z. Meng, Q. Yan, F. Zhao, and M. Liang, "Multi-scale feature attention and transformer for hyperspectral image classification," in *Proc. 13th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Oct./Nov. 2023, pp. 1–5.

[45] T. Zhang, S. Xiao, W. Dong, J. Qu, and Y. Yang, "A mutual guidance attention-based multi-level fusion network for hyperspectral and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[46] W. Dong, T. Zhang, J. Qu, S. Xiao, T. Zhang, and Y. Li, "Multibranch feature fusion network with Self- and cross-guided attention for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530612.

[47] Y. Zhang, T. Wang, C. Zhang, S. Xu, H. Gao, and C. Li, "A dual-branch Siamese spatial–spectral transformer attention network for hyperspectral image change detection," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 122125.

[48] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

[49] Y. Zhang et al., "Multimodal transformer network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514317.

[50] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213.

[51] Y. Zhang, Y. Peng, B. Tu, and Y. Liu, "Local information interaction transformer for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1130–1143, 2023.

[52] B. Tu, X. Liao, Q. Li, Y. Peng, and A. Plaza, "Local semantic feature aggregation-based transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536115.

[53] Y. Zhang et al., "A cross-modal feature aggregation and enhancement network for hyperspectral and LiDAR joint classification," *Expert Syst. Appl.*, vol. 258, Dec. 2024, Art. no. 125145.

[54] B. Tu, Q. Ren, J. Li, Z. Cao, Y. Chen, and A. Plaza, "NCGLF2: Network combining global and local features for fusion of multisource remote sensing data," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102192.

[55] L. Dong, W. Jiang, and J. Geng, "Hyperspectral and LiDAR data classification using spatial context and de-redundant fusion network," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[56] Y. Chu, M. Ye, and Y. Qian, "Incorporating attention mechanism and graph regularization into CNNs for hyperspectral image classification," in *Proc. 12th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2022, pp. 1–5.

[57] E. Khachatrian, S. Chlaily, T. Eltoft, and A. Marinoni, "A multimodal feature selection method for remote sensing data analysis based on double graph Laplacian diagonalization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11546–11566, 2021.

[58] J. Cai et al., "A novel graph-attention based multimodal fusion network for joint classification of hyperspectral image and LiDAR data," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123587.

[59] F. Guo, Z. Li, Q. Meng, L. Wang, and J. Zhang, "Dual graph convolution joint dense networks for hyperspectral and LiDAR data classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 1141–1144.

[60] Y. Cai, Z. Zhang, X. Liu, Y. Ding, F. Li, and J. Tan, "Learning unified anchor graph for joint clustering of hyperspectral and LiDAR data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 1, no. 1, pp. 1–14, May 2024.

[61] J. Feng, J. Zhang, and Y. Zhang, "Multi-view feature learning and multi-level information fusion for joint classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5528613.

[62] X. Liao, B. Tu, J. Li, and A. Plaza, "Class-wise graph embedding-based active learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522813.



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